A dyadic bound for torsion primes

27 October 2026

Abstract

We prove the bound $p \le \max\{(2^{d/2}+1)^2, 65(4d)^6\}$ for prime-order torsion on elliptic curves over number fields of degree $d \ge 3$. In particular, $p \le (2^{d/2}+1)^2$ for every $d \ge 53$. The argument combines Parent's modular-symbol estimate with Raynaud's bound on the failure of exactness of cotangent lattices at 2. Doubling a trace divisor moves it into a residue neighbourhood modulo 4, where the resulting map has a unique zero.

Let $S(d)$ be the set of primes occurring as orders of rational points on elliptic curves over number fields of degree d. Oesterlé's bound is $p \le (3^{d/2}+1)^2$ for $p \in S(d)$; see [3, Section 6]. We prove the following statement.

Theorem 1. *Let $d \ge 3$, let $K/\mathbb{Q}$ have degree d, and let E/K be an elliptic curve with a K-rational point of prime order p. Then*

$$p \le \max\bigl\{(2^{d/2}+1)^2,\ 65(4d)^6\bigr\}.$$

Consequently, for every $d \ge 53$, every prime divisor of $\#E(K)_{\mathrm{tors}}$ satisfies

$$p \le (2^{d/2}+1)^2.$$

We first record the local observations needed in the proof. All cotangent spaces below are relative to the indicated base ring.

Lemma 2. *Let f be a morphism between smooth $\mathbb{Z}_2$-schemes, let x_0 and y_0 be sections with $f(x_0) = y_0$, and choose coordinates $x_1, \dots, x_m$ for the formal neighbourhood of x_0. Suppose that the cotangent image of f at this section contains $2\mathbb{Z}_2^m$. Then the only point with all $x_i \in 4\mathbb{Z}_2$ mapping to y_0 is x_0.*

Proof. Choose integral formal functions on the target, vanishing at y_0, whose pullbacks are

$$G_i(x) = 2x_i + Q_i(x), \qquad 1 \le i \le m,$$

where each Q_i belongs to $(x_1, \dots, x_m)^2\mathbb{Z}_2[[x_1, \dots, x_m]]$. Suppose that $x \ne 0$ and put $k = \min_i v_2(x_i) \ge 2$. For an index attaining this minimum, the linear term of $G_i(x)$ has valuation $k+1$, whereas $Q_i(x)$ has valuation at least $2k > k+1$. Hence $G_i(x) \ne 0$. □

Lemma 3. *Let $X/\mathbb{Z}_2$ be a smooth curve with a section c, and let $D \in X^{(d)}(\mathbb{Z}_2)$ reduce to dc modulo 2. Then $2D \in X^{(2d)}(\mathbb{Z}_2)$ agrees with $2dc$ modulo 4 in the formal neighbourhood of $2dc$.*

Proof. Choose a parameter q along c. At dc, the elementary symmetric functions of the local parameters give coordinates on $X^{(d)}$. Encode their values by

$$E_D(t) = \prod_{j=1}^{d}(1 - q_j t) = 1 + 2B(t), \qquad B(t) \in \mathbb{Z}_2[t].$$

This expression is defined over $\mathbb{Z}_2$ even when the individual q_j are defined over ramified extensions or occur with repetitions. The corresponding polynomial for the doubled divisor is

$$E_{2D}(t) = E_D(t)^2 = 1 + 4\big(B(t) + B(t)^2\big).$$

Thus every nonconstant elementary symmetric coordinate of $2D$ is divisible by 4. □

For an abelian scheme $A/\mathbb{Z}_2$, the torsion in the kernel of $A(\mathbb{Q}_2) \to A(\mathbb{F}_2)$ is killed by 2. Indeed, multiplication by 2 sends the formal neighbourhood with coordinates in $2\mathbb{Z}_2$ into the neighbourhood with coordinates in $4\mathbb{Z}_2$. The latter is torsion free, either by the formal logarithm or the usual valuation argument for the multiplication series. We use this fact for the winding quotient.

Let p be an odd prime, put $J = J_0(p)$, and let $\mathbb{T}$ be the integral Hecke algebra acting on J. Let $\mathbf{e}$ be the winding element and put

$$I = \mathrm{Ann}_{\mathbb{T}}(\mathbf{e}), \qquad A = J/IJ, \qquad \pi : J \longrightarrow A.$$

The ideal I is saturated and π has connected kernel. The winding quotient $A(\mathbb{Q})$ is finite; see [4, Section 3.8]. Since p is odd, all these abelian varieties have good reduction at 2. Write

$$S = H^0(X_0(p)_{\mathbb{Z}_2}, \Omega^1), \qquad M = S[I], \qquad L = \pi^* \operatorname{Cot}_0(A_{\mathbb{Z}_2}) \subseteq S.$$

Lemma 4. *There are inclusions*

$$2M \subseteq L \subseteq M.$$

Moreover, the coefficient pairing gives an identification

$$M = \mathrm{Hom}_{\mathbb{Z}_2}\big((\mathbb{T}/I) \otimes \mathbb{Z}_2, \mathbb{Z}_2\big), \qquad f \longmapsto \big(T_n \longmapsto a_n(f)\big).$$

Proof. Put $B = \ker \pi$. Raynaud's theorem [1, Appendix A, Theorem A.1], applied to the exact sequence $0 \to B \to J \to A \to 0$ over $\mathbb{Q}_2$, gives

$$0 \longrightarrow \operatorname{Lie} B_{\mathbb{Z}_2} \longrightarrow \operatorname{Lie} J_{\mathbb{Z}_2} \longrightarrow \operatorname{Lie} A_{\mathbb{Z}_2} \longrightarrow (\mathbb{Z}/2\mathbb{Z})^r \longrightarrow 0$$

for some r. Here the ramification index is $1 = 2 - 1$, and the middle term has good reduction. Dualizing shows that the saturation of L in S contains L with quotient killed by 2; compare [2, Section 4.4, proof of Theorem 3.10]. The $\mathbb{Q}_2$-span of this saturation is the subspace annihilated by I. Hence its integral lattice is M, proving the inclusions.

The integral coefficient pairing between $\mathbb{T} \otimes \mathbb{Z}_2$ and S is perfect by [1, Lemma 2.1] and the integral q-expansion identification at primes not dividing the level. The submodule annihilated by I is consequently the dual of $(\mathbb{T}/I) \otimes \mathbb{Z}_2$. No Gorenstein hypothesis is needed. □

We next use the part of Parent's argument which concerns modular symbols. For $m \geq 3$, his combinatorial estimate implies that

$$T_i\{0, \infty\}, \qquad 1 \leq i \leq 2m,$$

are linearly independent in relative homology with coefficients in any field whenever $p > 65(2m)^6$; see [4, Proposition 1.9]. The corresponding result is Proposition 1.7 in the 1996 preprint. The following passage to winding symbols also works in characteristic 2.

Lemma 5. *If $m \geq 3$ and $p > 65(2m)^6$, then the map*

$$M \longrightarrow \mathbb{Z}_2^m, \qquad f \longmapsto (a_1(f), \dots, a_m(f))$$

is surjective. Consequently, the cotangent image at $m\infty$ of

$$f_m : X_0(p)^{(m)} \longrightarrow A, \qquad D \longmapsto \pi[D - m\infty],$$

contains $2\mathbb{Z}_2^m$.

Proof. We show that $T_1\mathbf{e}, \dots, T_m\mathbf{e}$ are independent modulo $2\mathbb{T}\mathbf{e}$. Put $\gamma = \{0, \infty\}$ and $h = T_2 - 3$. Both cusps are multiplied by 3 under T_2, so $h\gamma$ lies in integral absolute homology. The period definition of the winding element gives $h\gamma = h\mathbf{e}$. In particular, $h\mathbb{T}\mathbf{e}$ lies in integral absolute homology.

Suppose that

$$\sum_{i=1}^{m} \epsilon_i T_i \mathbf{e} \in 2\mathbb{T}\mathbf{e}, \qquad \epsilon_i \in \{0, 1\},$$

and let k be the largest index with $\epsilon_k = 1$. Multiplication by h gives a dependence modulo 2 between $T_i\gamma$ for $1 \leq i \leq 2k$, whose coefficient at $T_{2k}\gamma$ is 1. Indeed,

$$(T_2 - 3)T_i = T_{2i} + 2T_{i/2} - 3T_i,$$

where $T_{i/2} = 0$ for odd i. This contradicts Parent's estimate. This argument takes place in integral homology and uses no division by 2.

Since $\mathbb{T}\mathbf{e} \simeq \mathbb{T}/I$, independence modulo 2 makes the submodule generated by the first m operators primitive over $\mathbb{Z}_2$. Dualizing and applying Lemma 4 proves the coefficient surjectivity for M. The cotangent map of f_m in elementary symmetric coordinates is

$$\omega \longmapsto \big(a_1(\pi^*\omega), -a_2(\pi^*\omega), \dots, (-1)^{m-1}a_m(\pi^*\omega)\big);$$

see [4, Section 4.12]. The inclusion $2M \subseteq L$ proves the last assertion. □

Proof of Theorem 1. Suppose that p exceeds both terms in the maximum, and let $P \in E(K)$ have order p. At a place above 2 with residue degree $f \leq d$, good reduction would imply

$$p \leq \#\widetilde{E}(\mathbb{F}_{2^f}) \leq (2^{f/2} + 1)^2 \leq (2^{d/2} + 1)^2.$$

Additive reduction is impossible since its odd torsion has order at most 3. Nonsplit multiplicative reduction is also impossible: its odd torsion injects into a torus of order $2^f + 1$. Thus E has split multiplicative reduction at every place above 2. The reduction of P has order p in the component group, since $p > 2^f - 1$.

Let $x \in X_0(p)(K)$ be the image of $(E, \langle P \rangle)$ followed by the Atkin–Lehner involution. The component-group assertion says that x specializes to ∞ at every place above 2. Hence

$$D = \mathrm{Tr}_{K/\mathbb{Q}}(x) \in X_0(p)^{(d)}(\mathbb{Q})$$

reduces to $d\infty$ modulo 2. This statement allows repeated points when x is defined over a proper subfield of K.

The point $f_d(D) \in A(\mathbb{Q})$ is torsion and reduces to zero at 2. The reduction-kernel observation therefore gives

$$2f_d(D) = 0, \qquad f_{2d}(2D) = 0.$$

By Lemma 3, $2D$ lies in the residue neighbourhood of $2d\infty$ modulo 4. Put $m = 2d$. Our assumption $p > 65(4d)^6 = 65(2m)^6$ and Lemma 5 allow us to apply Lemma 2 to f_m. It follows that $2D = 2d\infty$, contradicting the noncuspidal support of D.

Finally,

$$65(4 \cdot 53)^6 = 5901039506984960 < 9007199254740992 = 2^{53}.$$

The ratio $2^d/d^6$ is increasing for integers $d \geq 53$, since $2(53/54)^6 > 1$ and $d/(d+1)$ is increasing. Thus $65(4d)^6 < 2^d < (2^{d/2}+1)^2$ for every $d \geq 53$. □

Remark 6. The theorem bounds torsion primes. It does not assert the same bound for the cardinality of the full torsion group or for arbitrary prime powers. The proof uses the exponent-two reduction-kernel statement and Raynaud's exponent-two cotangent defect. It does not require a uniform formal immersion theorem modulo 2, nor a uniform rank assertion for $J_0(p)$.

References

[1] A. Abbes and E. Ullmo, *À propos de la conjecture de Manin pour les courbes elliptiques modulaires*, Compositio Math. **103** (1996), 269–286. https://www.numdam.org/item/CM_1996__103_3_269_0/.

[2] A. Agashe, K. Ribet and W. A. Stein, *The Manin constant*, Pure Appl. Math. Q. **2** (2006), 617–636. https://wstein.org/papers/ars-manin/html/node12.html.

[3] M. Derickx, S. Kamienny, W. Stein and M. Stoll, *Torsion points on elliptic curves over number fields of small degree*, Algebra Number Theory **17** (2023), 267–308. https://arxiv.org/abs/1707.00364.

[4] P. Parent, *Bornes effectives pour la torsion des courbes elliptiques sur les corps de nombres*, J. Reine Angew. Math. **506** (1999), 85–116. Preprint: https://arxiv.org/abs/alg-geom/9611022.